MySQL Homework: Submission Checklist

Name: Sean Austin_________________________________

Individual Submission (as ZIP on Course Management System):

- SQL Scripts:
  - create.sql
  - insert.sql
  - drop.sql
  - queries.sql
- Documentation:
  - Database Schema
  - Database Population
  - For each non-trivial query:
    - Non-trivial Query
    - Reflection Query – Verifying correctness of Non-trivial Query
    - Comprehensive Test Data – Illustrating test cases with respect to the non-trivial query

All output must be legible.

Verified results should be indicated with colored ovals/text to show correctness.

In Word, Insert>Shapes>Oval (No Fill, Line Color).

Not recommended in google docs. If taking screen shots with snipit or snip & sketch tool utilize the drawing and highlight functions prior to inserting into document.

Database Schema

The describe command shows a description of the table's schema.

```
describe business;
describe business_categories;
describe user;
describe review;
```

business

Field	Type	Null	Key	Default	Extra
business_id	char(22)	NO	PRI	NULL	
name	varchar(100)	NO		NULL	
street	varchar(100)	NO		NULL	
city	varchar(50)	NO		NULL	
state	char(2)	NO		NULL	
zip	char(5)	NO		NULL	
stars	float	NO		NULL	
review_count	int	NO		NULL	

business_categories

Field	Type	Null	Key	Default	Extra
business_id	char(22)	NO	PRI	NULL	
category	varchar(50)	NO	PRI	NULL	

user

Field	Type	Null	Key	Default	Extra
user_id	char(22)	NO	PRI	NULL	
name	varchar(100)	NO		NULL	
review_count	int	NO		NULL	
yelping_since	char(10)	NO		NULL	

review

Field	Type	Null	Key	Default	Extra
review_id	char(22)	NO	PRI	NULL	
business_id	char(22)	NO	MUL	NULL	
user_id	char(22)	NO	MUL	NULL	
stars	int	NO		NULL	
review_date	char(10)	NO		NULL	
review_text	varchar(5000)	NO		NULL	

Database Population

The following illustrates the number of tuples in each table.

```
select count(*) as businessCount    FROM business;
select count(*) as categoryCount    FROM business_categories;
select count(*) as userCount        FROM user;
select count(*) as reviewCount      FROM review;
```

businessCount
10

categoryCount
20

userCount
10

reviewCount
37

Non-trivial Query 1

For each business, calculate the number of reviews and the average of the reviews using the data in the review table, and not the stored data for the business. Order results so that the business with the highest average review is listed first. So with that ordering, the business with the most reviews is listed first.
(business_id, name, business_review_count, avg_review)

Query

```
SELECT b.business_id, b.name, COUNT(r.review_id)  AS business_review_count,
AVG(r.stars)        AS avg_review
FROM business b
JOIN review r ON b.business_id = r.business_id
GROUP BY b.business_id, b.name
ORDER BY avg_review DESC, business_review_count DESC;
```

Answer:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

business_id	name	business_review_count	avg_review
biz0000000000000000005	Tacos Chiwas	4	5.0000
biz0000000000000000007	Matts Big Breakfast	3	4.6667
biz0000000000000000003	Changing Hands Bookstore	4	4.5000
biz0000000000000000001	Pizzeria Bianco	8	4.2500
biz0000000000000000008	Postino Arcadia	3	4.0000
biz0000000000000000010	Sip Coffee and Beer	2	4.0000
biz0000000000000000002	Flower Child	4	3.7500
biz0000000000000000006	The Lost Leaf	3	3.6667
biz0000000000000000004	Lux Central	4	3.0000
biz0000000000000000009	Ghost Ranch	2	3.0000

Reflection Query 1

Shows the raw review rows per business so the computed count and average can be manually verified.

Query

SELECT b.business_id, b.name, r.review_id, r.stars
FROM business b
JOIN review r ON b.business_id = r.business_id
ORDER BY b.business_id, r.review_id;

Answer:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

business_id	name	review_id	stars
biz000000000000000000001	Pizzeria Bianco	rev000000000000000000001	5
biz000000000000000000001	Pizzeria Bianco	rev000000000000000000002	4
biz000000000000000000001	Pizzeria Bianco	rev000000000000000000003	4
biz000000000000000000001	Pizzeria Bianco	rev000000000000000000004	4
biz000000000000000000001	Pizzeria Bianco	rev000000000000000000005	5
biz000000000000000000001	Pizzeria Bianco	rev000000000000000000006	4
biz000000000000000000001	Pizzeria Bianco	rev000000000000000000007	4
biz000000000000000000001	Pizzeria Bianco	rev000000000000000000008	4
biz000000000000000000002	Flower Child	rev000000000000000000009	4
biz000000000000000000002	Flower Child	rev000000000000000000010	4
biz000000000000000000002	Flower Child	rev000000000000000000011	4
biz000000000000000000002	Flower Child	rev000000000000000000012	3
biz000000000000000000003	Changing Han...	rev000000000000000000013	5
biz000000000000000000003	Changing Han...	rev000000000000000000014	4
biz000000000000000000003	Changing Han...	rev000000000000000000015	5
biz000000000000000000003	Changing Han...	rev000000000000000000016	4
biz000000000000000000004	Lux Central	rev000000000000000000017	3
biz000000000000000000004	Lux Central	rev000000000000000000018	4
biz000000000000000000004	Lux Central	rev000000000000000000019	3
biz000000000000000000004	Lux Central	rev000000000000000000020	2
biz000000000000000000005	Tacos Chiwas	rev000000000000000000021	5
biz000000000000000000005	Tacos Chiwas	rev000000000000000000022	5
biz000000000000000000005	Tacos Chiwas	rev000000000000000000023	5
biz000000000000000000005	Tacos Chiwas	rev000000000000000000024	5
biz000000000000000000006	The Lost Leaf	rev000000000000000000025	4
biz000000000000000000006	The Lost Leaf	rev000000000000000000026	4
biz000000000000000000006	The Lost Leaf	rev000000000000000000027	3
biz000000000000000000007	Matts Big Bre...	rev000000000000000000028	4
biz000000000000000000007	Matts Big Bre...	rev000000000000000000029	5
biz000000000000000000007	Matts Big Bre...	rev000000000000000000030	5
biz000000000000000000008	Postino Arcadia	rev000000000000000000031	4
biz000000000000000000008	Postino Arcadia	rev000000000000000000032	4
biz000000000000000000008	Postino Arcadia	rev000000000000000000033	4
biz000000000000000000009	Ghost Ranch	rev000000000000000000034	3
biz000000000000000000009	Ghost Ranch	rev000000000000000000035	3
biz000000000000000000010	Sip Coffee an...	rev000000000000000000036	4
biz000000000000000000010	Sip Coffee an...	rev000000000000000000037	4

Comprehensive Test Data 1

Shows business review count, and the min/max stars, and average to confirm the query handles all edge cases.

Query

```sql
SELECT b.business_id, b.name,
   COUNT(r.review_id)  AS review_count,
   MIN(r.stars)        AS min_stars,
   MAX(r.stars)        AS max_stars,
   AVG(r.stars)        AS avg_stars
FROM business b
JOIN review r ON b.business_id = r.business_id
GROUP BY b.business_id, b.name
ORDER BY avg_stars DESC, review_count DESC;
```

Answer:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

business_id	name	review_count	min_stars	max_stars	avg_stars
biz0000000000000000005	Tacos Chiwas	4	5	5	5.0000
biz0000000000000000007	Matts Big Breakfast	3	4	5	4.6667
biz0000000000000000003	Changing Hands Bookstore	4	4	5	4.5000
biz0000000000000000001	Pizzeria Bianco	8	4	5	4.2500
biz0000000000000000008	Postino Arcadia	3	4	4	4.0000
biz0000000000000000010	Sip Coffee and Beer	2	4	4	4.0000
biz0000000000000000002	Flower Child	4	3	4	3.7500
biz0000000000000000006	The Lost Leaf	3	3	4	3.6667
biz0000000000000000004	Lux Central	4	2	4	3.0000
biz0000000000000000009	Ghost Ranch	2	3	3	3.0000

Test data illustrates:

- biz5 (Tacos Chiwas): 4 reviews all 5-star → avg 5.0, ranks first
- biz7 (Matts Big Breakfast): 3 reviews → avg 4.67, ranks second
- biz4 vs biz9: both avg 3.0, biz4 has 4 reviews vs biz9 has 2 → biz4 ranks higher (but they tie break by count)
- biz10 (Sip Coffee and Beer): 2 reviews minimum count, verifies no business is excluded

Non-trivial Query 2

For each business category in the database, calculate the number of different businesses, the number of reviews, and the average review for that category. Order the results alphabetically by category.
(category, number_of_businesses, category_review_count, category_average_review)

Query

```
SELECT bc.category,
   COUNT(DISTINCT bc.business_id)  AS number_of_businesses,
   COUNT(r.review_id)              AS category_review_count,
   AVG(r.stars)                    AS category_average_review
FROM business_categories bc
JOIN review r ON bc.business_id = r.business_id
GROUP BY bc.category
ORDER BY bc.category ASC;
```

Answer:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	number_of_businesses	category_review_count	category_average_review
Bars	2	5	3.8000
Bookstores	1	4	4.5000
Breakfast & Brunch	1	3	4.6667
Cafes	1	4	3.0000
Coffee & Tea	2	6	3.3333
Mexican	2	6	4.3333
Music Venues	1	3	3.6667
Pizza	1	8	4.2500
Restaurants	6	24	4.2083
Salad	1	4	3.7500
Shopping	1	4	4.5000
Wine Bars	1	3	4.0000

Reflection Query 2

Shows individual businesses and their reviews per category so the aggregated totals can be manually verified.

Query

```
SELECT bc.category, bc.business_id, b.name, r.review_id, r.stars
FROM business_categories bc
JOIN business b ON bc.business_id = b.business_id
JOIN review   r ON bc.business_id = r.business_id
ORDER BY bc.category, bc.business_id, r.review_id;
```

Answer:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	business_id	name	review_id	stars
Bars	biz0000000000000000006	The Lost Leaf	rev0000000000000000025	4
Bars	biz0000000000000000006	The Lost Leaf	rev0000000000000000026	4
Bars	biz0000000000000000006	The Lost Leaf	rev0000000000000000027	3
Bars	biz0000000000000000010	Sip Coffee and Beer	rev0000000000000000036	4
Bars	biz0000000000000000010	Sip Coffee and Beer	rev0000000000000000037	4
Bookstores	biz0000000000000000003	Changing Hands Bookstore	rev0000000000000000013	5
Bookstores	biz0000000000000000003	Changing Hands Bookstore	rev0000000000000000014	4
Bookstores	biz0000000000000000003	Changing Hands Bookstore	rev0000000000000000015	5
Bookstores	biz0000000000000000003	Changing Hands Bookstore	rev0000000000000000016	4
Breakfast & Brunch	biz0000000000000000007	Matts Big Breakfast	rev0000000000000000028	4
Breakfast & Brunch	biz0000000000000000007	Matts Big Breakfast	rev0000000000000000029	5
Breakfast & Brunch	biz0000000000000000007	Matts Big Breakfast	rev0000000000000000030	5
Cafes	biz0000000000000000004	Lux Central	rev0000000000000000017	3
Cafes	biz0000000000000000004	Lux Central	rev0000000000000000018	4
Cafes	biz0000000000000000004	Lux Central	rev0000000000000000019	3
Cafes	biz0000000000000000004	Lux Central	rev0000000000000000020	2
Coffee & Tea	biz0000000000000000004	Lux Central	rev0000000000000000017	3
Coffee & Tea	biz0000000000000000004	Lux Central	rev0000000000000000018	4
Coffee & Tea	biz0000000000000000004	Lux Central	rev0000000000000000019	3
Coffee & Tea	biz0000000000000000004	Lux Central	rev0000000000000000020	2
Coffee & Tea	biz0000000000000000010	Sip Coffee and Beer	rev0000000000000000036	4
Coffee & Tea	biz0000000000000000010	Sip Coffee and Beer	rev0000000000000000037	4
Mexican	biz0000000000000000005	Tacos Chiwas	rev0000000000000000021	5
Mexican	biz0000000000000000005	Tacos Chiwas	rev0000000000000000022	5
Mexican	biz0000000000000000005	Tacos Chiwas	rev0000000000000000023	5
Mexican	biz0000000000000000005	Tacos Chiwas	rev0000000000000000024	5
Mexican	biz0000000000000000009	Ghost Ranch	rev0000000000000000034	3
Mexican	biz0000000000000000009	Ghost Ranch	rev0000000000000000035	3
Music Venues	biz0000000000000000006	The Lost Leaf	rev0000000000000000025	4
Music Venues	biz0000000000000000006	The Lost Leaf	rev0000000000000000026	4
Music Venues	biz0000000000000000006	The Lost Leaf	rev0000000000000000027	3
Pizza	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000001	5
Pizza	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000002	4
Pizza	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000003	4
Pizza	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000004	4
Pizza	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000005	5
Pizza	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000006	4

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	business_id	name	review_id	stars
Pizza	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000008	4
Restaurants	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000001	5
Restaurants	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000002	4
Restaurants	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000003	4
Restaurants	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000004	4
Restaurants	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000005	5
Restaurants	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000006	4
Restaurants	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000007	4
Restaurants	biz0000000000000000001	Pizzeria Bianco	rev0000000000000000008	4
Restaurants	biz0000000000000000002	Flower Child	rev0000000000000000009	4
Restaurants	biz0000000000000000002	Flower Child	rev0000000000000000010	4
Restaurants	biz0000000000000000002	Flower Child	rev0000000000000000011	4
Restaurants	biz0000000000000000002	Flower Child	rev0000000000000000012	3
Restaurants	biz0000000000000000005	Tacos Chiwas	rev0000000000000000021	5
Restaurants	biz0000000000000000005	Tacos Chiwas	rev0000000000000000022	5
Restaurants	biz0000000000000000005	Tacos Chiwas	rev0000000000000000023	5
Restaurants	biz0000000000000000005	Tacos Chiwas	rev0000000000000000024	5
Restaurants	biz0000000000000000007	Matts Big Breakfast	rev0000000000000000028	4
Restaurants	biz0000000000000000007	Matts Big Breakfast	rev0000000000000000029	5
Restaurants	biz0000000000000000007	Matts Big Breakfast	rev0000000000000000030	5
Restaurants	biz0000000000000000008	Postino Arcadia	rev0000000000000000031	4
Restaurants	biz0000000000000000008	Postino Arcadia	rev0000000000000000032	4
Restaurants	biz0000000000000000008	Postino Arcadia	rev0000000000000000033	4
Restaurants	biz0000000000000000009	Ghost Ranch	rev0000000000000000034	3
Restaurants	biz0000000000000000009	Ghost Ranch	rev0000000000000000035	3
Salad	biz0000000000000000002	Flower Child	rev0000000000000000009	4
Salad	biz0000000000000000002	Flower Child	rev0000000000000000010	4
Salad	biz0000000000000000002	Flower Child	rev0000000000000000011	4
Salad	biz0000000000000000002	Flower Child	rev0000000000000000012	3
Shopping	biz0000000000000000003	Changing Hands Bookstore	rev0000000000000000013	5
Shopping	biz0000000000000000003	Changing Hands Bookstore	rev0000000000000000014	4
Shopping	biz0000000000000000003	Changing Hands Bookstore	rev0000000000000000015	5
Shopping	biz0000000000000000003	Changing Hands Bookstore	rev0000000000000000016	4
Wine Bars	biz0000000000000000008	Postino Arcadia	rev0000000000000000031	4
Wine Bars	biz0000000000000000008	Postino Arcadia	rev0000000000000000032	4
Wine Bars	biz0000000000000000008	Postino Arcadia	rev0000000000000000033	4

Comprehensive Test Data 2

Confirms the test data covers: a category with only one business, categories shared by multiple businesses, and businesses belonging to multiple categories.

Query

```
SELECT bc.category,
    COUNT(DISTINCT bc.business_id)  AS num_businesses,
    COUNT(r.review_id)              AS total_reviews,
    AVG(r.stars)                    AS avg_stars
FROM business_categories bc
JOIN review r ON bc.business_id = r.business_id
GROUP BY bc.category
ORDER BY num_businesses DESC, bc.category;
```

Answer:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	num_businesses	total_reviews	avg_stars
Restaurants	6	24	4.2083
Bars	2	5	3.8000
Coffee & Tea	2	6	3.3333
Mexican	2	6	4.3333
Bookstores	1	4	4.5000
Breakfast & Brunch	1	3	4.6667
Cafes	1	4	3.0000
Music Venues	1	3	3.6667
Pizza	1	8	4.2500
Salad	1	4	3.7500
Shopping	1	4	4.5000
Wine Bars	1	3	4.0000

Test data illustrates:

- Restaurants: shared by 6 businesses largest multi-business category
- Mexican: shared by biz5 (Tacos Chiwas) and biz9 (Ghost Ranch)
- Pizza, Salad, Bookstores, etc: single-business categories verifies 1 business case
- biz10 in both Coffee & Tea and Bars verifies reviews count toward each category independently